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(54) **WET PET FOOD DISPENSER**

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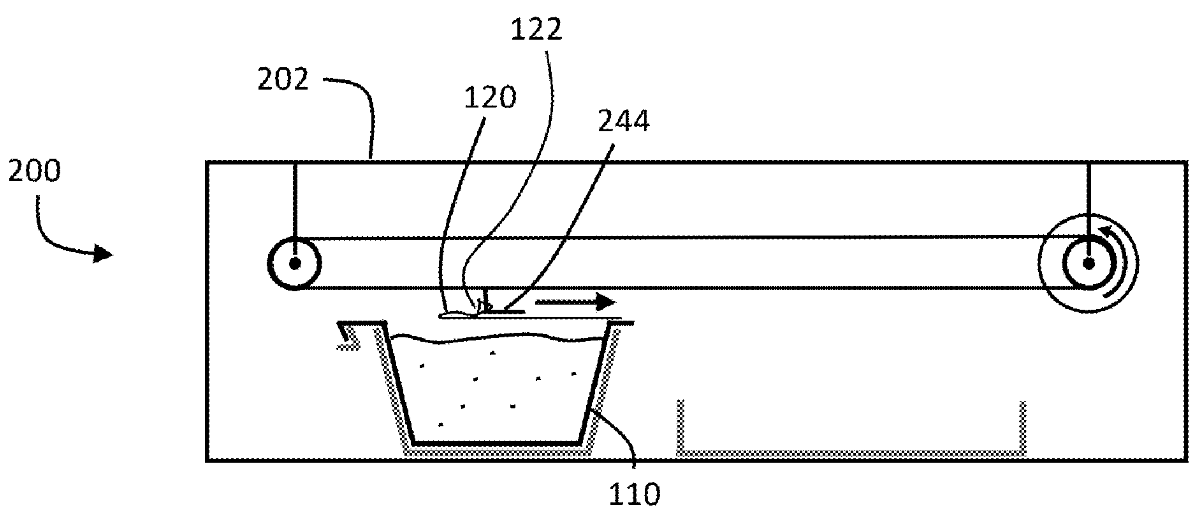
(57) **ABSTRACT**

In one aspect, a device for automatically serving wet pet food includes a body and a lid removal bay in the body configured to receive a pet food pod. The pod includes a container portion containing wet pet food. The pod is generally bowl-shaped and has a rim. The pod further includes a lid, peelably and hermetically sealed to the rim, having a peripheral tab. The device further includes a tab engaging element configured to engage the tab and a lid removal actuator operable to actuate the tab engaging element along a trajectory over a pet food pod in the lid removal bay so that the tab engaging element will engage and pull the tab across the container portion to cause the lid to peel away from the rim. A retainer is configured to substantially immobilize the pod within the lid removal bay during lid removal.



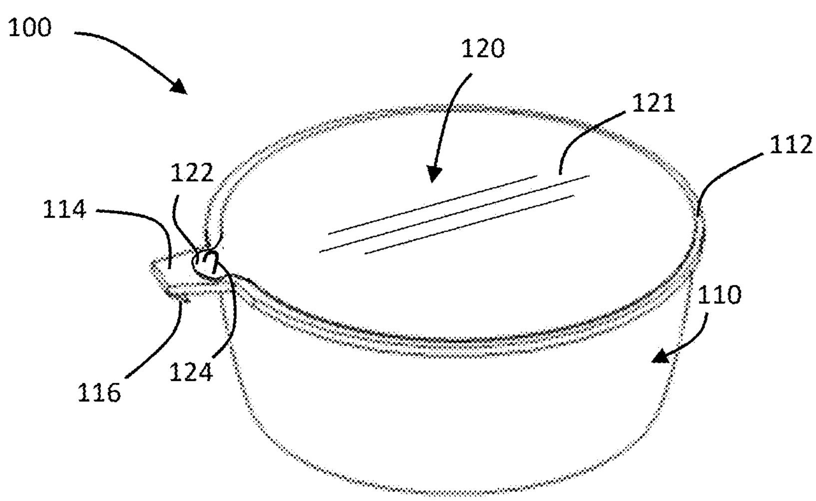


FIG. 1

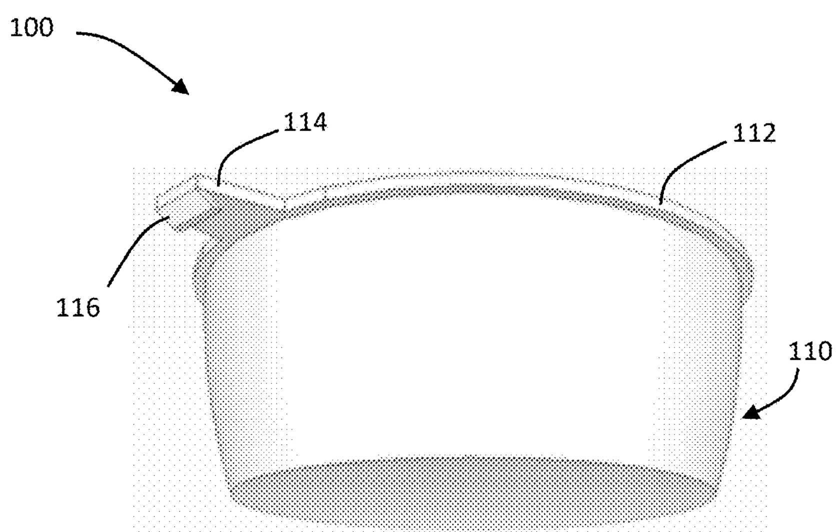


FIG. 2

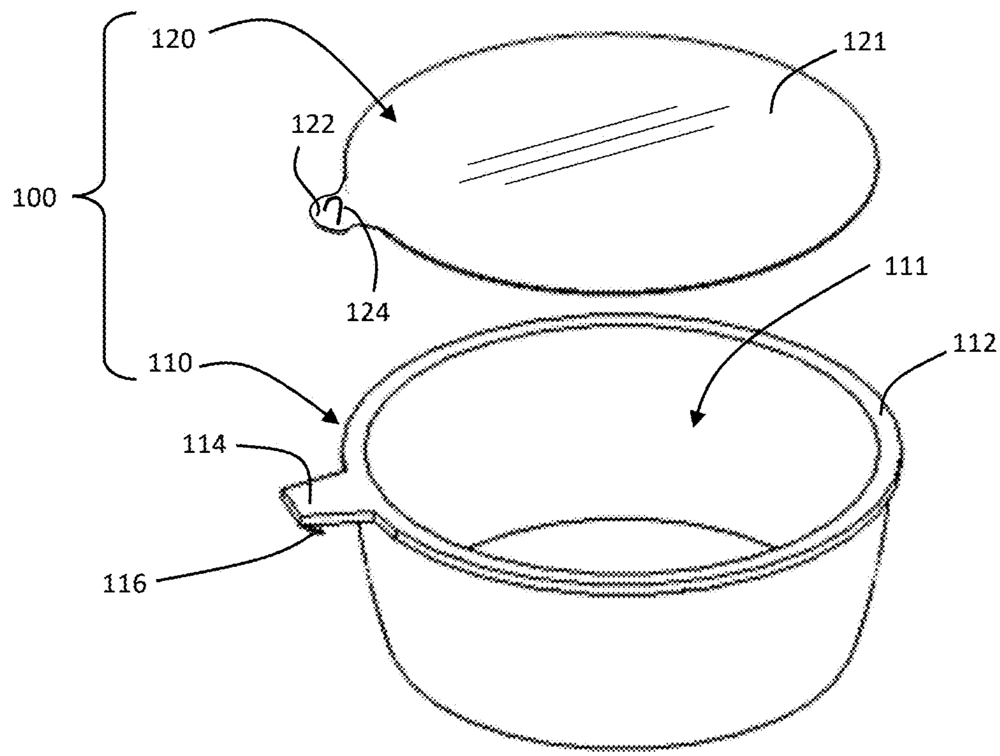


FIG. 3

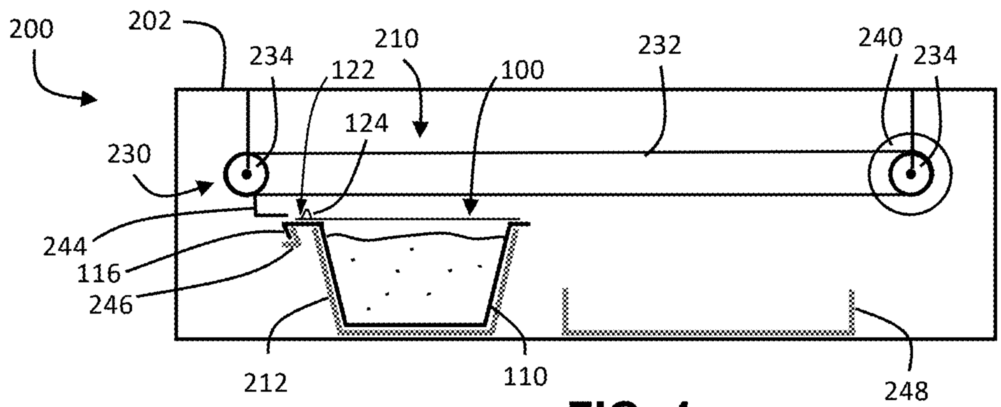


FIG. 4

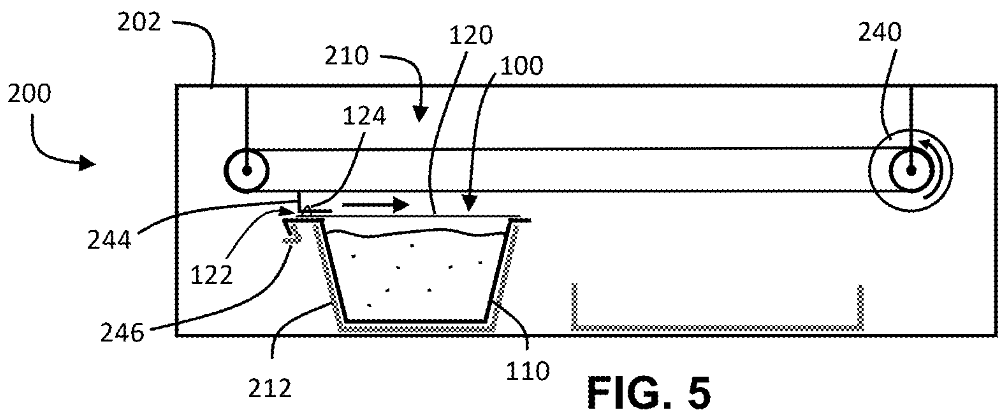


FIG. 5

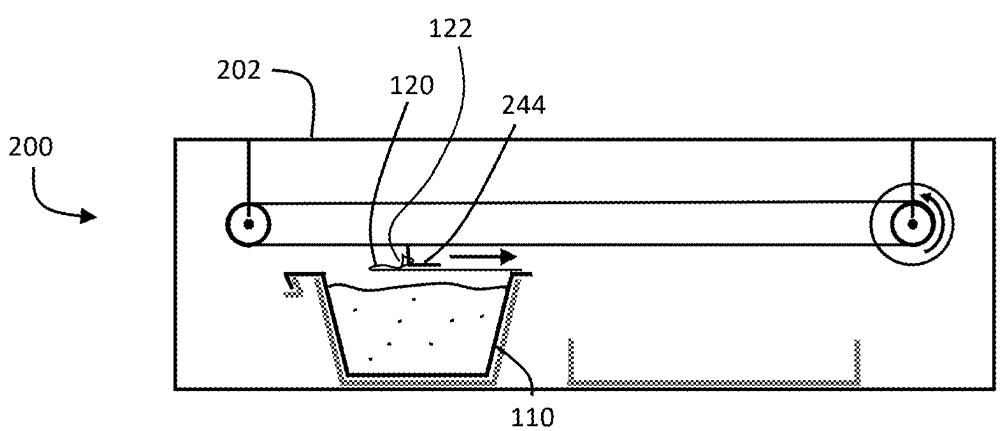


FIG. 6

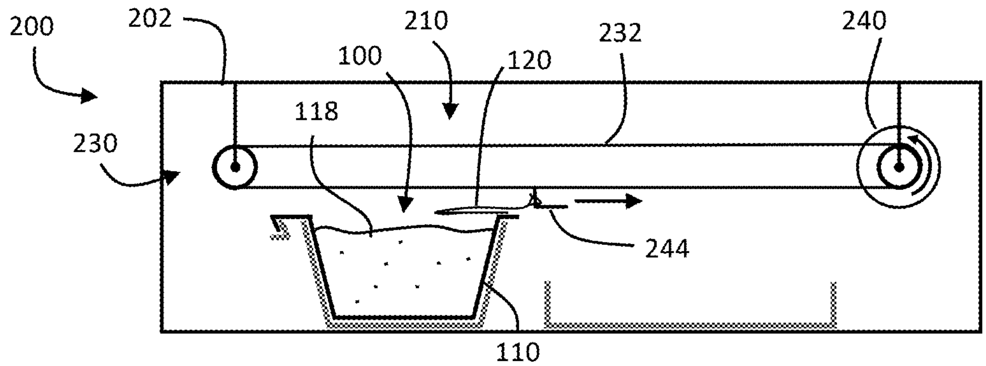


FIG. 7

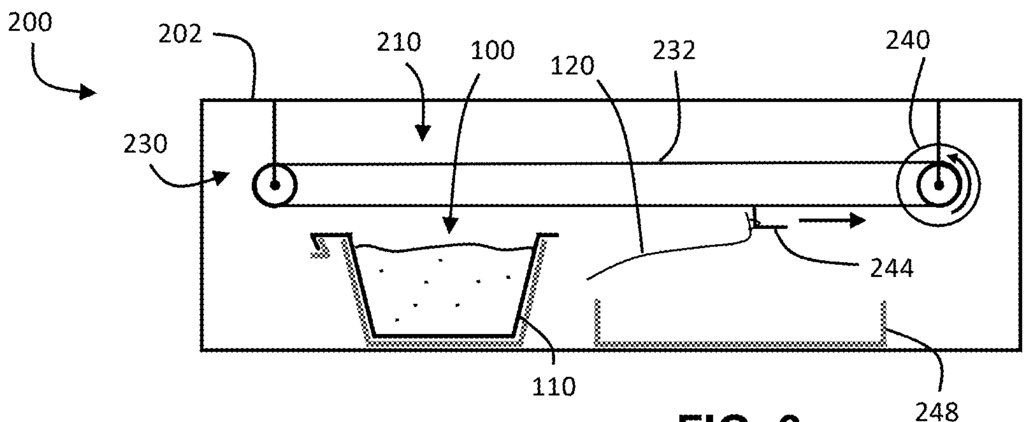


FIG. 8

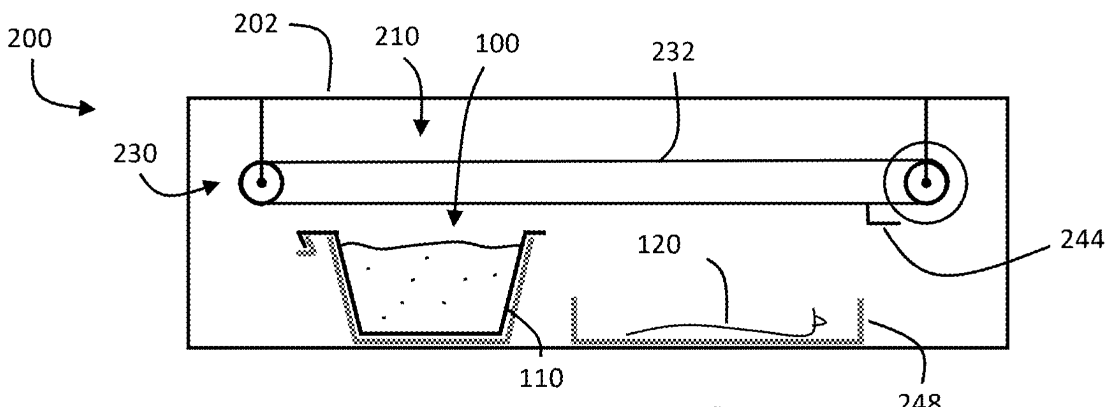


FIG. 9

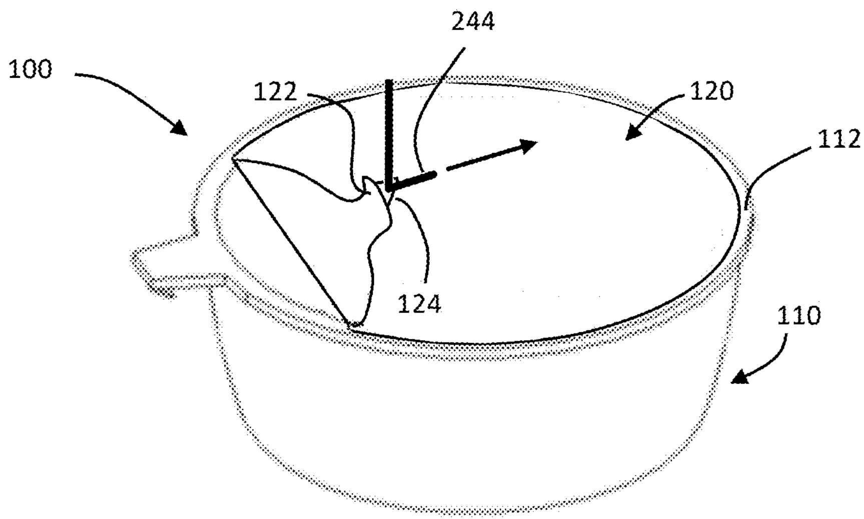


FIG. 10

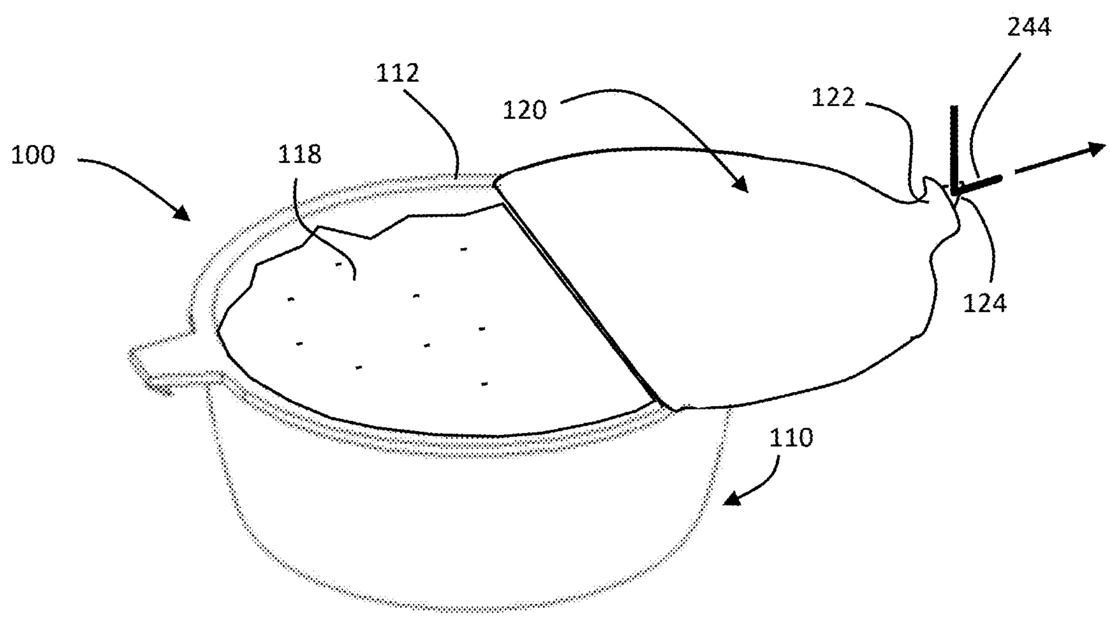


FIG. 11

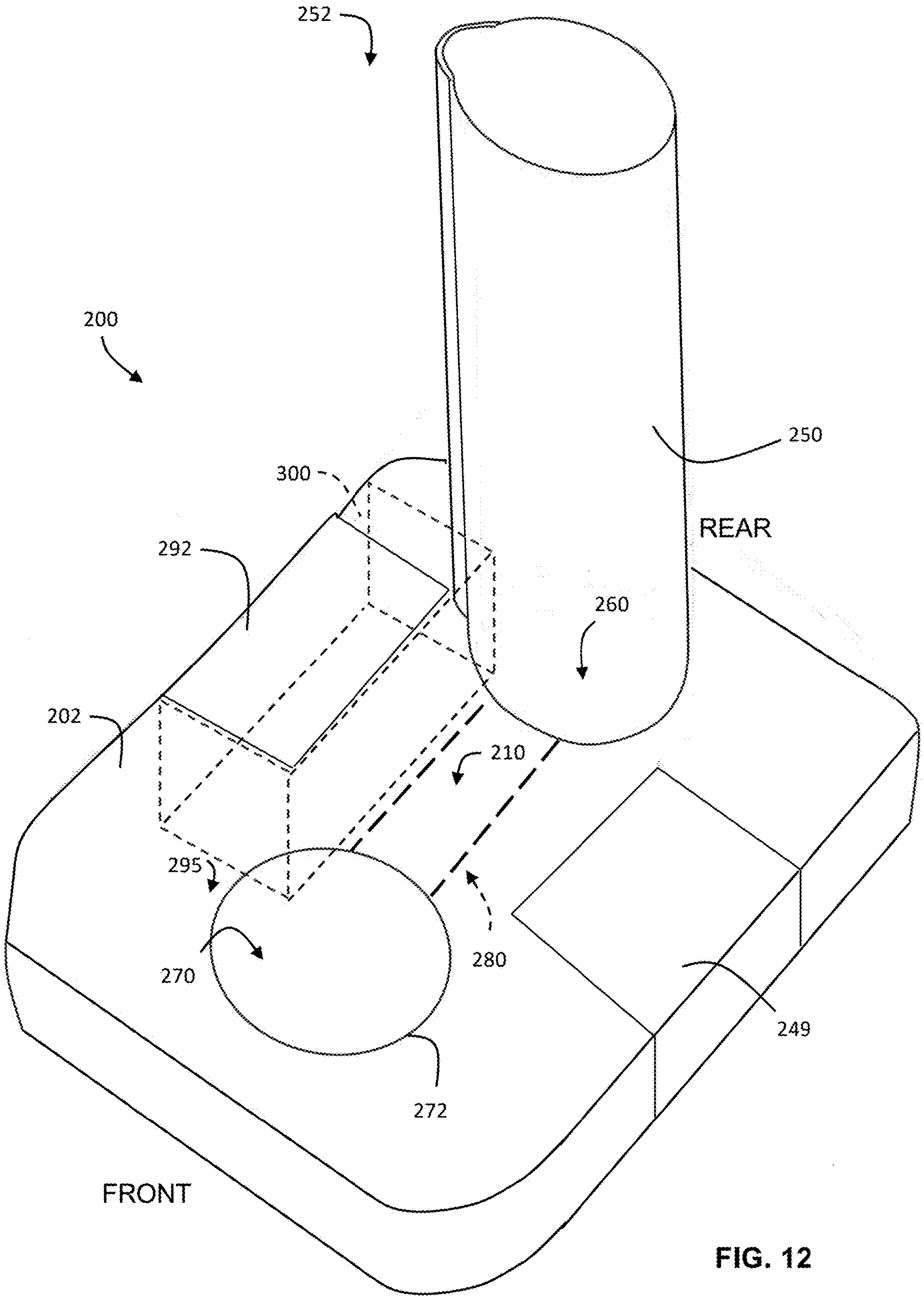


FIG. 12

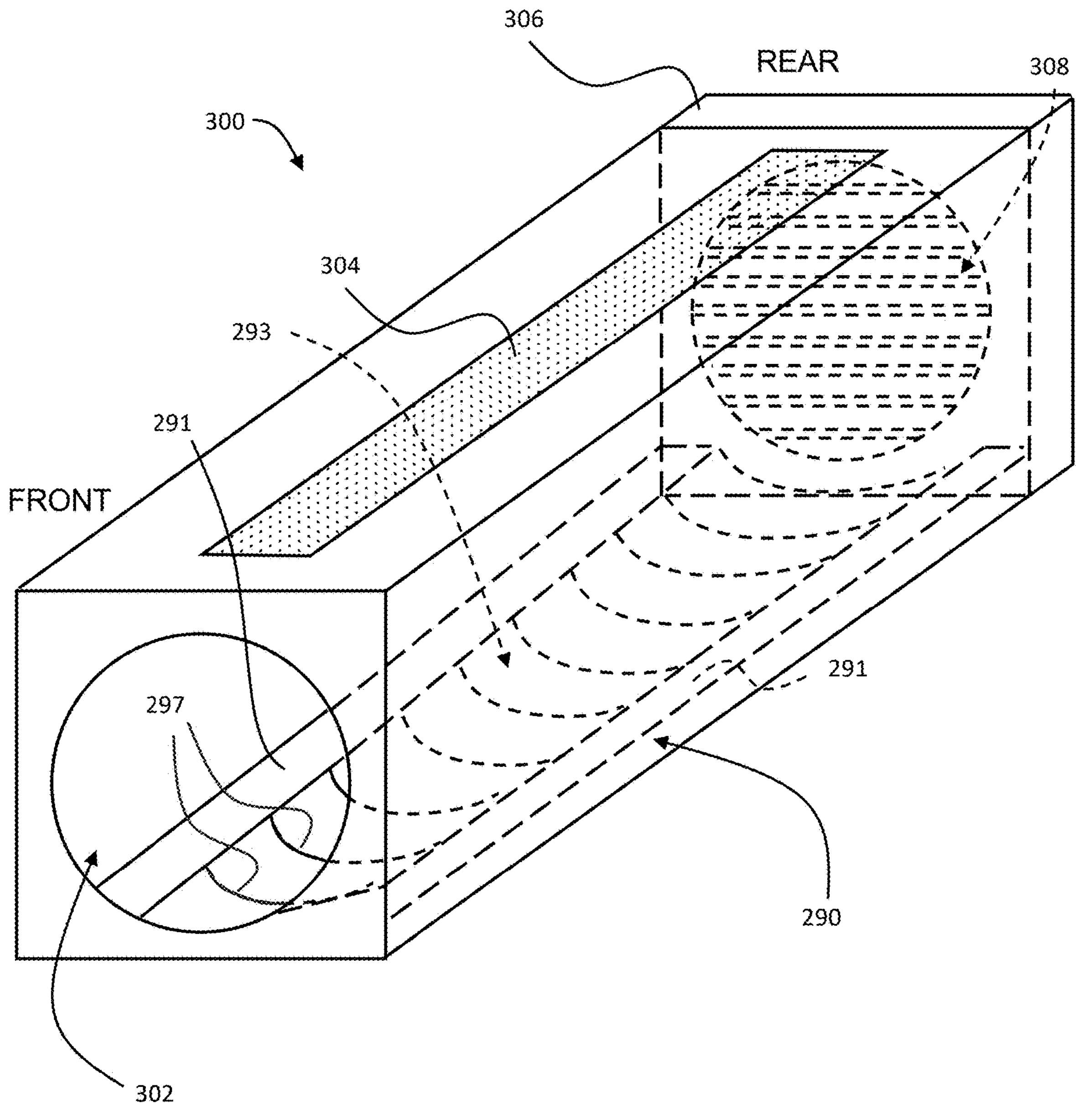
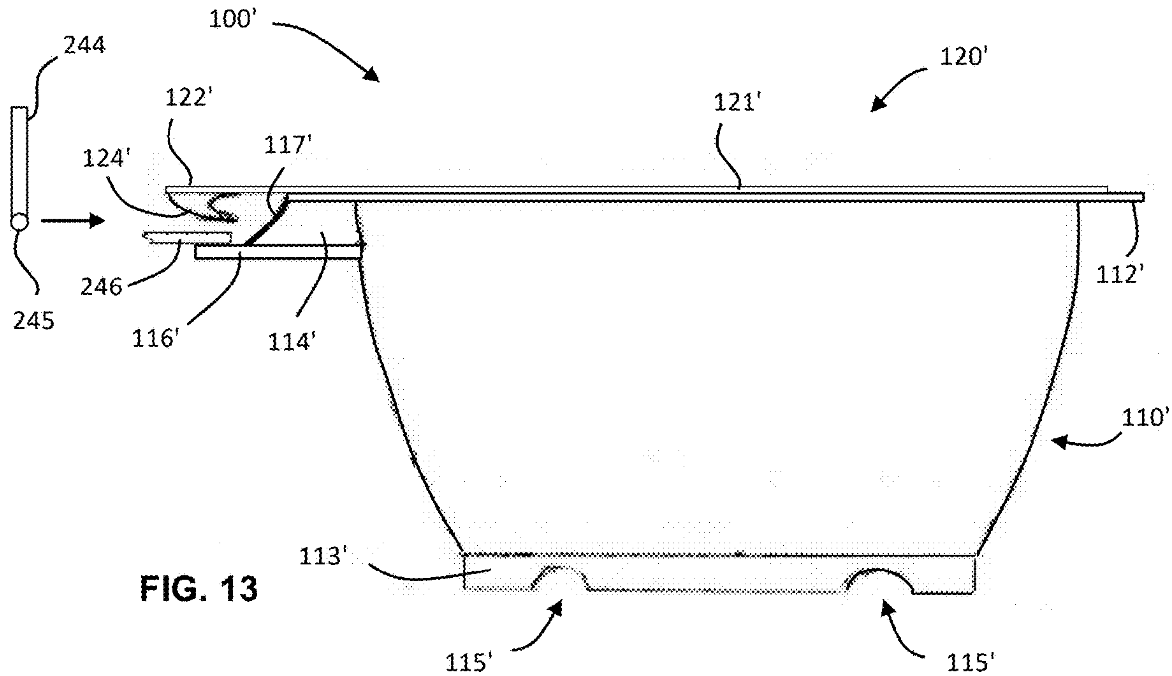
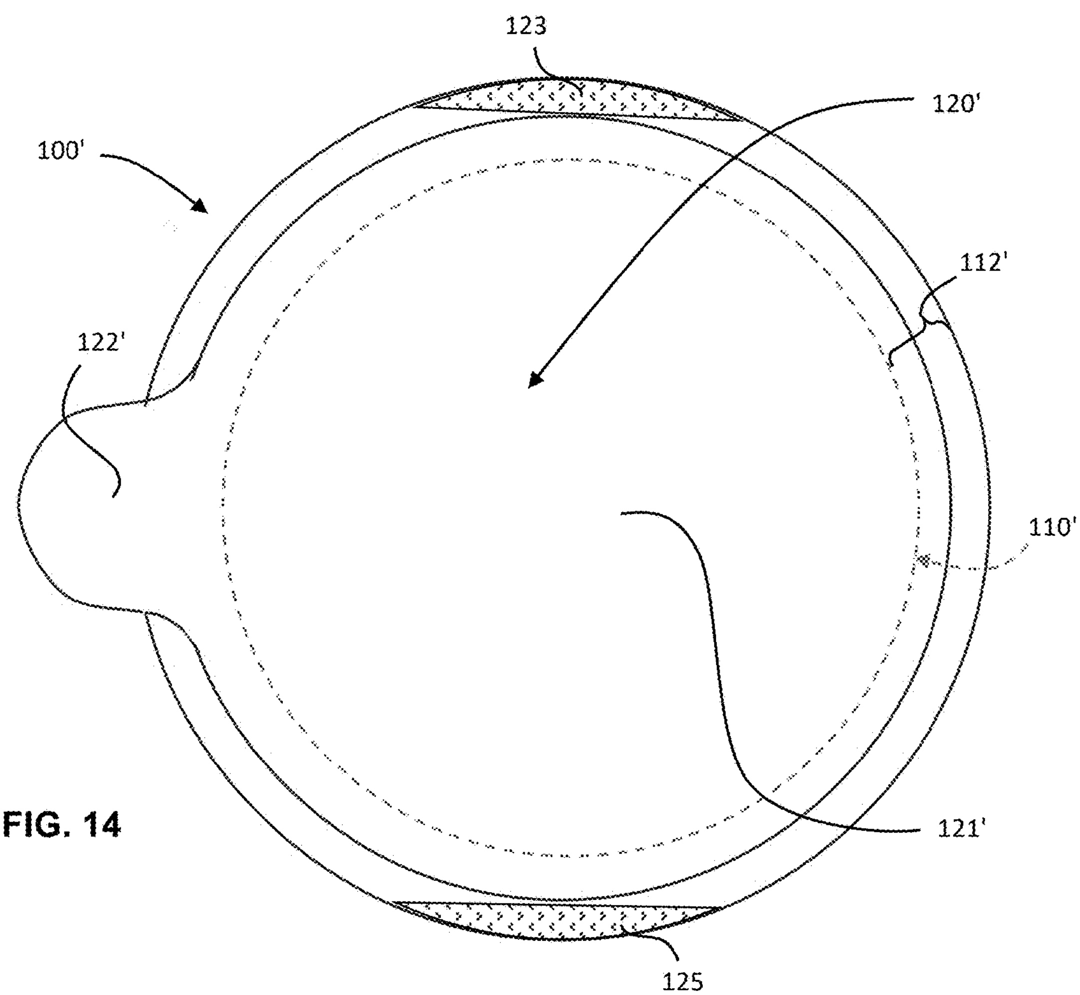


FIG. 12A



244
100'
120'
122'
121'
124'
117'
245
246
116'
114'
112'
110'
113'
115'
115'

FIG. 13



123
120'
100'
112'
122'
110'
121'
125

FIG. 14

WET PET FOOD DISPENSER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of U.S. Provisional Patent Application No. 63/557,242 filed on Feb. 23, 2024, which is hereby incorporated by reference hereinto.

TECHNICAL FIELD

[0002] The present disclosure relates to pet food, and more particularly to a device for automatically serving wet pet food.

BACKGROUND

[0003] Dry food for house pets, such cats or dogs, is shelf-stable and convenient. However, in some respects, dry pet food may not be as healthy for pets as wet pet food. For example, eating solely dry pet food may leave some cats insufficiently hydrated because cats may lack a strong thirst drive. Pets may also find dry pet food to be less appetizing than wet pet food. This may lead to pet undernourishment.

SUMMARY

[0004] In one aspect of the present disclosure, there is provided a device for automatically serving wet pet food, the device comprising: a body; a lid removal bay in the body, the lid removal bay configured to receive a pet food pod, the pet food pod including a container portion containing wet pet food, the container portion being generally bowl-shaped and having a rim, the pet food pod further including a lid peelably and hermetically sealed to the rim, the lid having a peripheral tab; a tab engaging element configured to engage the peripheral tab; a lid removal actuator in the body, the lid removal actuator operable to, upon receipt of the pet food pod within the lid removal bay, actuate the tab engaging element along a trajectory over the pet food pod so that the tab engaging element will engage and pull the tab across the container portion to cause the lid to peel away from the rim, resulting in an open container portion with exposed wet pet food; and a retainer configured to substantially immobilize the pet food pod within the lid removal bay during lid removal by the tab engaging element.

[0005] In some embodiments, the device further comprises a pod storage tower removably attached to the body, the pod storage tower for receiving a stack of the pet food pods.

[0006] In some embodiments, the pod storage tower includes a pod alignment feature for promoting a uniform orientation of the pet food pods in the stack received in the pod storage tower.

[0007] In some embodiments, the device further comprises a receiving bay in the body below the pod storage tower, the receiving bay configured to receive a pet food pod from the pod storage tower; and a pod lowering mechanism for lowering a lowermost pod from the pod storage tower into the receiving bay.

[0008] In some embodiments, the device further comprises a conveyor operable to convey a pet food pod from the receiving bay to the lid removal bay.

[0009] In some embodiments, the conveyor substantially maintains an orientation of the pet food pod during conveyance.

[0010] In some embodiments, the conveyor comprises a pair of parallel rails, the container portion of the pet food pod has a base with a pair of straight parallel grooves formed in an underside of the base, and the conveyor is operable to slidably translate the pet food pod atop the pair of parallel rails with each of the grooves seated on a respective one of the rails.

[0011] In some embodiments, the device further comprises a lid disposal tray within the body adjacent to the lid removal bay, the lid disposal tray being configured to hold a plurality of lids after removal of the lids from respective ones of the pet food pods.

[0012] In some embodiments, the device further comprises a feeding bay in the body configured to receive the open container portion of a pet food pod that has been conveyed from the lid removal bay, the feeding bay including an opening in a top surface of the body, the opening being surrounded by an inwardly facing lip, the lip being configured to obstruct pod removal from the feeding bay without obstructing access to wet pet food within the container portion.

[0013] In some embodiments, the device further includes a pod disposal chamber configured to receive a plurality of pet food pod container portions, in a substantially empty state, conveyed from the feeding bay.

[0014] In some embodiments, the device further includes a compactor mechanism operable to compact the plurality of pet food pod container portions, in the substantially empty state, within the pod disposal chamber.

[0015] In some embodiments, the pod disposal chamber includes a removable pod disposal tray having a pair of longitudinal curbs and a floor between the curbs recessed downwardly relative to the curbs.

[0016] In some embodiments, the pod disposal chamber includes a removable pod disposal tray having a floor and a plurality of crosswise grooves or ridges on the floor, each of the crosswise grooves or ridges being configured to catch the rim of a respective one of the pet food pod container portions to discourage slippage of the pet food pod container portion towards a front of the pod disposal tray.

[0017] In some embodiments, the device further comprises a fan within the body; and a removable filter positioned in an airflow path of the fan, the removable filter configured to absorb or neutralize odors from air circulated by the fan, the filter being removably secured within the device.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] In the figures which illustrate example embodiments:

[0019] FIG. **1** is a top perspective view of an example pet food pod embodiment;

[0020] FIG. **2** is a bottom perspective view of the example pet food pod of FIG. **1**;

[0021] FIG. **3** is an exploded view of the pet food pod of FIGS. **1** and **2**;

[0022] FIGS. **4**, **5**, **6**, **7**, **8**, and **9** are schematic views of a device for automatically serving wet pet food at various stages of operation;

[0023] FIG. **10** depicts an example pet food pod with a partially opened lid as it may appear when the device is at the stage of operation depicted in FIG. **6**;

[0024] FIG. **11** depicts an example pet food pod with a mostly opened lid as it may appear when the device is at the stage of operation depicted in FIG. **7**;

[0025] FIG. **12** is a front top right perspective view of an embodiment of a device for automatically serving wet pet food;

[0026] FIG. **12**A is a schematic view of part of the device of FIG. **12**;

[0027] FIG. **13** is a side elevation view of an alternative embodiment of pet food pod that may be loaded into the device embodiment of FIG. **12**; and

[0028] FIG. **14** is a top view of the pet food pod of FIG. **13**.

DESCRIPTION

[0029] The present disclosure describes a device for automatically serving wet pet food to a house pet. The device is operable to automatically open one or more sealed pods containing wet pet food that has/have been loaded into the device.

[0030] FIGS. **1** and **2** are a top perspective view and bottom perspective view, respectively, of a pet food pod **100** containing wet pet food. FIG. **3** is an exploded view of the pet food pod **100** of FIGS. **1** and **2** in which the wet pet food is omitted for clarity. The pet food pod **100** has a container portion **110** and a lid **120**.

[0031] The container portion **110** is configured (e.g., shaped and sized) to contain a single serving of wet pet food that, with the lid **120** removed, can be eaten by a pet directly from the container portion **110**. The container portion **110** is generally bowl-shaped with an open top **111**, best seen in FIG. **3**. The container portion **110** is made from a moisture-impermeable material, such as plastic. The container portion **110** has a rim **112**, which in the present embodiment is an outwardly projecting annular flange having a flat upper surface.

[0032] A protrusion **114** extends laterally from the rim **112** on one side of the pet food pod **100**. The protrusion **114** has a pod retention element **116**. As will be appreciated, the pod retention element **116** may facilitate immobilization of the pet food pod **100** during removal of the lid **120**. In the present embodiment, the pod retention element **116** is a lip (tab) that extends downwardly and inwardly from a distal end of the protrusion **114** (see, e.g., FIG. **2**).

[0033] The lid **120** is formed of a moisture-impermeable material, such as foil or a foil material with a plastic undercoating. In the present embodiment, the lid **120** has a generally circular body portion **121** and an integral tab **122** extending from a periphery of the body portion **121** (i.e., a peripheral tab **122**). The peripheral tab **122** is used during lid removal as described below. In the present embodiment, the peripheral tab **122** has an upstanding loop **124** of strong material, such as wire, nylon thread, or plastic filament, securely attached thereto. The loop **124** facilitates engagement of the tab **122** during lid removal and may thus be generically referred to as a tab engagement facilitating portion **124** of the tab **122**. The material from which the loop **124** is made may be pliable or rigid and may have a sufficient degree of stiffness for loop **124** to be upstanding as shown in FIG. **1** for example. The upstanding profile of loop **124** may facilitate engagement of the peripheral tab **122** by some implementations of lid removal mechanisms, as will be described.

[0034] The pet food pod **100** may be sold with wet pet food contained in the container portion **110** and the lid **120** peelably and hermetically sealed to the rim **112** to keep the pet food fresh and moist. The lid **120** may be sealed to the rim **112** using a food-safe adhesive, which is not expressly depicted. The adhesive may be applied in a continuous ring that seals a periphery of the body portion **121** of the lid **120** to the upper surface of rim **112** but leaves the tab **122** unadhered to the protrusion **114** with which the tab **122** is aligned. Leaving the tab **122** unadhered to the container portion **110** may facilitate the initiation of lid removal, as will be described. The extent of the body portion **121** of the lid **120** may be less than the extent of the rim **112**, i.e., the rim **112** may protrude outwardly peripherally from underneath the body portion **121** of lid **120**. In some embodiments, the protruding portion of the rim **112** may facilitate stabilizing the pod **100** during lid removal, e.g., via one or more elements that engage with (hold down) the rim **112** without inhibiting removal of the lid **120**, as will be described. The degree of protrusion of rim **112** from underneath the body portion **121** of lid **120** may vary and may be greater than what is shown in FIG. **1** for example.

[0035] A device **200** for automatically serving wet pet food is schematically depicted in FIGS. **4-9** at various stages of operation. The device **200** is configured to receive at least one pet food pod **100** as depicted in FIGS. **1-3** (or pet food pod **100'** as depicted in FIGS. **13** and **14**, described below) and to automatically open the pet food pod **100** for presentation to a house pet for consumption, e.g., at a time when the owner of the pet is away from home.

[0036] Referring to FIG. **4**, the schematically depicted device **200** has a body **202** made from a rigid material, such as metal or plastic. A lid removal bay **210** is disposed within the body **202**. The lid removal bay **210** is configured to receive a pet food pod **100**, e.g., in a receptacle **212** that may be fixed with respect to the body **202**.

[0037] Also disposed within the body **202** is a lid removal mechanism **230**. The lid removal mechanism **230** is designed to facilitate automatic removal of the lid **120** of a pet food pod **100** that has been received in the lid removal bay **210**. In the present embodiment, the lid removal mechanism **230** includes an endless drive member **232** (e.g., a chain or belt) looped around a pair of rotatable gears or pulleys **234**. A lid removal actuator **240** (e.g., an electric motor) is operable to induce rotary motion of one of the gears or pulleys **234**, causing the endless drive member **232** to loop around both gears or pulleys **234**. A tab engaging element **244**, which in this embodiment is an L-shaped hook, is attached to the endless drive member **232** as depicted.

[0038] The device **200** also includes a retainer **246**. The retainer **246** is a physical component that is configured to substantially immobilize the pet food pod **100** within the lid removal bay **210** during removal of the lid **120** by the lid removal mechanism **230**. The retainer **246** may have a complementary shape to that of the pod retention element **116** for engagement therewith and may be fixed with respect to the body **202**. In this embodiment, the retainer **246** has an upstanding lip (or tab) that is angled away from the receptacle **212** at a complementary angle to that of the pod retention element **116**.

[0039] The example device **200** further includes a lid disposal tray **248** adjacent to the lid removal bay **210**. The lid disposal tray **248** is configured to receive at least one lid **120** that has been removed from a pet food pod **100**. The lid disposal tray **248** may be removable to facilitate disposal of one or more lids **120** contained therein.

[0040] The device **200** may optionally include other elements and components which are omitted from FIGS. **4-9** for simplicity and clarity. Examples of other elements and components that may be included in the device **200** are described below.

[0041] In operation, the lid removal bay **210** receives a pet food pod **100** in receptacle **212** (see FIG. **4**). The pet food pod **100** may be provided to the lid removal bay **210** in various ways. In one embodiment, a conveyor (not depicted) may convey the pet food pod **100** to the lid removal bay **210** from another area of device **200**, such as a receiving bay below a pod storage tower, both of which are described below. Alternatively, the pet food pod **100** may be placed into the lid removal bay **210** manually in some embodiments, e.g., via an access door (not depicted) in the body **202** of the device **200**.

[0042] When the pet food pod **100** is received in the receptacle **212**, the retainer **246** engages the pod retention element **116**, e.g., in an interlocking manner. Within the lid removal bay **210**, the pet food pod **100** is oriented so that the protrusion **114**, and thus tab **122** and loop **124**, are aligned with the tab engaging element **244** of the lid removal mechanism **230**.

[0043] Referring to FIG. **5**, the lid removal actuator **240** of the lid removal mechanism **230** may be automatically activated, e.g., by a controller (not depicted) of the device **200**, e.g., at a preset time of day, which may be user configurable. In this embodiment, activation of the actuator **240** causes the endless drive member **232** to rotate, which in turn causes tab engaging element **244** to begin to move (translate) to the right in FIG. **5**. The tab engaging element **244** moves along a trajectory over the pet food pod **100** in which it will engage with the tab **122**. More specifically, in this embodiment, the L-shaped hook of tab engaging element **244** engages with the tab **122** by passing through the tab engagement facilitating portion **124** (i.e., upstanding loop **124**) of tab **122**, as shown in FIG. **5**.

[0044] Thereafter, the tab engaging element **244** continues to move along the trajectory defined by the endless drive member **232**. In so doing, the tab engaging element **244** pulls tab **122**, by way of loop **124**, across the container portion **110** of the pet food pod **100**. The retainer **246** helps to keep the pet food pod **100** immobilized within the lid removal bay **210** during lid removal by virtue of its engagement with the pod retention element **116**. In some embodiments, immobilization of the pet food pod **100** may be further promoted by one or more elements that engage with a protruding portion of rim **112** (see FIG. **1**). For example, rails (not depicted) forming part of the body **202** may be positioned on opposite sides of the pet food pod **100** directly over the protruding portion of rim **112**, e.g., with each rail being offset approximately ninety degrees from the protrusion **114**.

[0045] The force generated by the actuator **240** is sufficient for the pulling of tab **122** to cause the lid **120** to start to peel away from the rim **112** of the container portion **110**. The lid folds over itself as it is peeled away, e.g., as shown in FIG. **6**. A perspective view of the partially opened state of the pet food pod **100** at this stage of operation is provided in FIG. **10**.

[0046] Referring to FIG. **7**, continued actuation of the tab engaging element **244** along the trajectory continues to peel away the lid **120** from the rim **112** of the container portion **110**, exposing the wet pet food **118** within the container portion **110**. A perspective view of the mostly opened state of the pet food pod **100** at this stage of operation is provided in FIG. **11**.

[0047] Referring to FIG. **8**, continued actuation of the tab engaging element **244** by the lid removal actuator **240** causes the lid **120** to peel fully away (i.e., to be removed) from the container portion **110**. The detached lid **120** may be carried away from the lid removal bay **210** towards the lid disposal tray **248** by the tab engaging element **244**.

[0048] Referring to FIG. **9**, the lid **120** has been dropped into the lid disposal tray **248** for eventual disposal. The mechanism that causes the lid **120** to be dropped may vary between embodiments. In one embodiment, the mechanism may cause a downward angling of tab engaging element **244** so that the loop **124** is dropped by the operation of gravity. In another embodiment, the loop **124** may be severed, e.g., by a static blade (not depicted).

[0049] The fully opened pet food pod **100** may thereafter be presented to a pet for consumption of the wet pet food **118**. This may be done by conveying the container portion **110** to a feeding bay of the device **200**, as described below. Alternatively, a panel or door (not depicted) in the body **202** of the device **200** may be automatically opened to provide access to the wet pet food **118**.

[0050] A front top right perspective view of one embodiment of device **200** is provided in FIG. **12**. The device **200** depicted in FIG. **12** may be used with an embodiment of pet food pod **100'** that is depicted in side elevation view in FIG. **13** and in top view in FIG. **14**.

[0051] As shown in FIGS. **13** and **14**, the pet food pod **100'** has a container portion **110'** and a lid **120'**. The container portion **110'** is generally bowl-shaped and has rim **112'**. In the present embodiment, the rim **112'** is an outwardly projecting annular flange having a flat upper surface. A base **113'** of the container portion **110'** has two parallel grooves **115'** (e.g., channels) therein, which are shown end-on in FIG. **13**. Each groove **115'** is straight and extends from one side of the base **113'** to the other.

[0052] A protrusion **114'** (e.g., a nub) extends laterally from the pet food pod **100'** on one side. The protrusion **114'** has an inwardly and upwardly sloped lateral face **117'**, which has a curved profile in the present embodiment (but not necessarily in other embodiments). As will be appreciated, the face **117'** may facilitate engagement of a tab engaging element **244** of device **200** with a hook **124'** forming part of the pet food pod **100'** (described below).

[0053] The protrusion **114'** has an associated pod retention element **116'**. The pod retention element **116'** may be used to facilitate immobilization of the pet food pod **100'** during removal of the lid **120'**. In the present embodiment, the pod retention element **116'** is a flat ledge that extends laterally from a wall of the container portion **110'** immediately below protrusion **114'**. The protrusion **114'** and pod retention element **116'** are both oriented substantially perpendicularly with respect to grooves **115'** (i.e., to the dimension in which the grooves **115'** are oriented).

[0054] The lid **120'** is formed of a moisture-impermeable material, such as foil or a foil material with a plastic undercoating. In the present embodiment, the lid **120'** has a generally circular body portion **121'** and an integral tab **122'** extending from a periphery of the body portion **121'** (i.e., a peripheral tab **122'**). In the present embodiment, an inwardly and downwardly pointing hook **124'** is securely attached to an underside of tab **122'**. The hook **124'**, which may be curved and may, e.g., have a shape resembling that of a dorsal fin of a dolphin, can be used to facilitate lid removal as described below. As such, the hook **124'** may generically be referred to as a tab engagement facilitating portion **124'** of tab **122'**.

[0055] The lid **120'** is peelably and hermetically sealed to the rim **112'** of the container portion **110'** to seal in the wet pet food (not expressly depicted). The lid **120'** may be sealed to the rim **112'** using a food-safe adhesive, which is not expressly depicted. The adhesive may be applied in a continuous ring that seals a periphery of the body portion **121'** of the lid **120'** to the upper surface of rim **112'**. The extent of the lid **120'** may be less than an extent of rim **112'**, i.e., the rim **112'** may protrude outwardly peripherally from underneath the body portion **121'** of lid **120'**.

[0056] Referring to FIG. **12**, the device has a body **202** made from a rigid material. A pod storage tower **250** sits atop the body **202**, protruding upwardly from to the body **202**. In this embodiment, the pod storage tower **250** is attached to an upper surface of the body **202** near a rear of the device **200**. The pod storage tower **250** may be removably attached to the body **202** to facilitate cleaning of the tower **250** and/or body **202**.

[0057] The pod storage tower **250** is configured to loosely receive a stack of pet food pods **100'**. In the depicted embodiment, the pod storage tower **250** is generally cylindrical. In some embodiments, the pod storage tower **250** is made of a transparent material to allow a user to see the number of pet food pods **100'** stacked therein and to visually verify that the pods **100'** are not inadvertently flipped or otherwise misaligned so as to risk jamming device **200**.

[0058] The pod storage tower **250** may have a pod alignment feature **252**. In this embodiment, the pod alignment feature **252** is an outdent configured (e.g., sized and shaped) to accommodate loosely therewithin the tab **122'**, protrusion **114'**, and pod retention element **116'** of the pet food pod **100'** (see FIGS. **13** and **14**). The pod alignment feature **252** may promote a uniform orientation of pet food pods **100'** that are stacked inside the pod storage tower **250**.

[0059] In some embodiments, the pod storage tower **250** has a top hatch to selectively permit pet food pods **100'** to be loaded from the top. In some embodiments, a latchable side or rear door (not depicted) may be situated at or near the bottom of the pod storage tower **250**. The latter door may permit manipulation of the pet food pods **100'** therewithin to correct any possible pod flipping or jamming issues. In some embodiments, the lower door may have a width equivalent to less than 100% (e.g., approximately 30%) of a diameter of the pod storage tower **250**. This may be sufficient for a user to insert one or more fingers into the pod storage tower **250** to correct any possible pod jamming issues without risking spillage of any of the pet food pods **100** out through the door.

[0060] A receiving bay **260** may be disposed within the body below the pod storage tower **250**. The receiving bay **260** may be configured to receive a pet food pod from the pod storage tower **250**. The receiving bay **260** may include a pod lowering mechanism (not expressly depicted) for lowering a single pet food pod **100** at the lowest position of the pod storage tower **250** into the receiving bay **260**. Operation of such a pod lowering mechanism is described below.

[0061] The lid removal bay **210**, described above, is disposed within the body **202**, e.g., adjacent to the receiving bay. In the present embodiment, the lid removal bay **210** is located substantially centrally within the body **202**. The lid removal bay **210** is configured to receive a pet food pod **100** conveyed from the receiving bay **260**. The lid removal bay **210** has an associated retainer **246**, described above.

[0062] The lid disposal tray **248**, described above, is located in the body **202** adjacent to the lid removal bay **210**, which is at the right side of FIG. **12** in the present embodiment. A latchable lid removal access door **249** may permit the lid disposal tray **248** to be accessed and removed as necessary, e.g., for emptying and/or cleaning.

[0063] The lid removal mechanism **230**, described above, is disposed within the body near the lid removal bay **210** and the lid disposal tray **248**. The lid removal mechanism **230** is configured to actuate a tab engaging element **244**, which in this embodiment is an L-shaped hook, along a trajectory from the lid removal mechanism **230** to the lid disposal tray **248**.

[0064] The device **200** may further include a feeding bay **270** at which an opened pet food pod **100'** may be presented to a pet for feeding. In the present embodiment, the feeding bay **270** is located near the front of the device **200**. The feeding bay **270** may include an opening in the body **202** that is surrounded by an inwardly facing lip **272**. The opening may be circular. The lip **272** may be configured to cover the rim **112'** of the opened pet food pod **100'** to prevent pod removal during feeding. The lip does not obstruct pet access to the wet pet food **118**.

[0065] The device **200** may further include a conveyor **280** within the body **202**. The conveyor **280** may be operable to convey a pet food pod **100'** from the receiving bay **260** to the lid removal bay **210**, and in some embodiments, subsequently from the lid removal bay **210** to the feeding bay **270**. The conveyor **280** may include a pair of rails (not expressly depicted) along which the pet food pod **100'** may be conveyed, e.g., between the receiving bay **260** and the lid removal bay **210**. The rails may be spaced apart suitably for alignment with the grooves **115'** on the underside of a pet food pod **100'**. Seating of the grooves **115'** on the rails may facilitate the maintaining of a desired orientation of the pet food pod **100'** as it is translated along the rails. Maintaining a particular orientation of the pet food pod **100'** may promote a desired interaction with other device components such as the retainer **246** and the lid removal mechanism **230**, e.g., as described above.

[0066] The device **200** may further include a removable pod disposal tray **290**. The purpose of the pod disposal tray **290** is to hold one or more used pet food pods **100'**. For clarity, the term "used pet food pods" herein refers to substantially empty pet food pod container portions after the wet pet food **118** therein has been substantially consumed by a pet. The pod disposal tray **290** may be part of a pod disposal chamber **300** within the body **202** of the device **200**. The pod disposal chamber **300** (and thus the pod disposal tray **290**) may be located on an opposite side of the body **202** from the lid disposal tray **248**, as shown in FIG. **12**. An example pod disposal chamber **300** is schematically depicted in FIG. **12A** in isolation.

[0067] Referring to FIG. **12A**, the pod disposal chamber **300** of the present embodiment is a cuboid chamber that receives and stores used pet food pod containers. In the depicted embodiment, the pod disposal chamber **300** has an opening **302** at its front end—which may be oriented toward the front of the body **202** of the device **200**—through which the used pods may be received.

[0068] The pod disposal tray **290** may be located at the bottom of the pod disposal chamber **300**. The pod disposal tray **290** may be defined by a floor **293** flanked by a pair of longitudinal curbs **291**. In the depicted embodiment, the floor **293** is recessed downwardly relative to the curbs **291**, i.e., is lower than the curbs **291**. The floor **293** of the present embodiment is curved in the transverse dimension. In some embodiments, the floor **293** may be sloped upwardly towards a rear of the pod disposal chamber **300**. In some embodiments, the floor **293** may have crosswise (transverse) grooves **297** or ridges along its length, as depicted in FIG. **12**A, for reasons that will be described below. In some embodiments, the pod disposal tray **290** may be a tray with walls and a floor.

[0069] The example pod disposal chamber **300** of FIG. **12**A includes a disinfecting light source **304**. The light source is operable to emit light, such as ultraviolet light, into an interior of the chamber **300** to facilitate sanitizing of the chamber **300** and the surfaces of any pods contained therewithin.

[0070] The example pod disposal chamber **300** further includes a fan unit **306** at the rear of the pod disposal chamber **300** to facilitate ventilation thereof. The fan unit **306** includes an electric fan (not expressly depicted) adjacent to a vented opening **308**. A replaceable air filter (not expressly depicted) may be insertable into a receptacle adjacent to the vented opening **308**, which may be accessible through an opening in body **202**. The filter may be removably secured within the device, e.g., using a friction fit, a clipping mechanism, or a fastener, such as one or more snaps, screws, magnets, or hook-and-loop fasteners. The air filter may be positioned in an airflow path of the fan and may incorporate an agent, such as baking soda and/or activated carbon, to facilitate absorption and/or neutralization of odors.

[0071] A latchable door **292** (FIG. **12**) may selectively provide user access to the pod disposal chamber **300** for removal of pod disposal tray **290**, e.g., for emptying and/or cleaning. The latch may be designed so that opening by a house pet is difficult or impossible. A second conveyor **295** may be configured to convey an empty or substantially empty pet food pod **100'** from the feeding bay **270** to the pod disposal tray **290**.

[0072] The device **200** may operate as follows, e.g., automatically under the control of a controller (not expressly depicted). At a predetermined feeding time, the pod lowering mechanism may retrieve a lowermost pet food pod **100'** from the pod storage tower **250**. In some embodiments, the pod lowering mechanism may include an opposed pair of curved arms (not depicted). The curved arms may be operable to hug and hold the pet food pod **100'** from opposite sides and to lower the pet food pod **100** gently into the receiving bay **260**. Lowering may be preferable to freefall dropping of the pet food pod **100** to reduce a risk of pod misorientation within the receiving bay **260**. The curved arms may be configured to move apart to disengage the pet food pod **100** when the pet food pod **100** has been lowered into the receiving bay **260**. The arms may thereafter move back up in preparation for lowering the next pet food pod **100** in the future.

[0073] By virtue of the pod alignment feature **252** of the pod storage tower **250**, the orientation of the lowered pet food pod **100'** will be such that the grooves **115'** in the base **113'** of the pet food pod **100'** align with (e.g., receive respective ones of) the rails of the conveyor **280**. The relative sizing of the grooves **115'** relative to the rails may provide for a margin of error for possible misalignment as the pod **100'** is lowered onto the rails. For example, the rails may be thinner underneath the pod storage tower **250** and may increase in thickness towards the lid removal bay **210**. The rails may be considered as a form of track.

[0074] In a subsequent operation, the controller may cause the pet food pod **100'** to be conveyed along the track from the receiving bay **260** to the lid removal bay **210**. More specifically, the pet food pod **100'** may be conveyed (translated) along the rails by two arms situated alongside (e.g., on opposite side of) the track that may be used to push pod **100** along the rails of the conveyor **280**. The arms may be curved to receive respective opposed portions of the pod **100'**, i.e., to cradle the pod **100'** from opposite side. The widening rails may correct any minor initial misalignment of the pet food pod **100'** as it is conveyed towards and into the lid removal bay **210**. This may reduce any possible shifting, and may promote proper alignment, of the pet food pod **100'** upon being received at the lid removal bay **210**.

[0075] At the lid removal bay **210**, the pet food pod **100'** may be substantially immobilized as follows. The pod retention element **116'** may be positioned in alignment with a retainer **246** forming part of device **200**. For the embodiment of pod **100'** depicted in FIGS. **13** and **14**, the retainer **246** may be a shelf-like structure in fixed relation to the body **202** of the device **200**. The retainer **246** may be sized and positioned so as to occupy a space immediately above pod retention element **116'**, e.g., as shown in FIG. **13**, when the pod **100'** is in position within the lid removal bay **210**. The retainer **246** may be positioned just above the pod retention element **116'** so as to substantially limit or prevent lifting up of the pod retention element **116'** during lid removal. In the embodiment depicted in FIG. **13**, there is a space (gap) above the retainer **246** and below tab engagement facilitating portion (hook) **124'** that is sufficiently large to receive the portion **245** of tab engaging element **244** during lid removal.

[0076] In some embodiments, additional elements of the device **200**, e.g., a fixed pair of rails (not depicted), may engage with other portions of the pod **100'** to help substantially immobilize the pod **100'** in the lid removal bay **210** during lid removal. For example, the rails may be positioned so as to overhang the areas **123**, **125** of the upper surface of the protruding portion of rim **112'** on opposite sides of the pod **100'** immediately above rim **112'**. This may further help to limit or prevent upward lifting of the container portion **110'** during lid removal.

[0077] Thereafter, the controller may activate the lid removal mechanism **230** to initiate removal of the lid **120'**. Operation of the lid removal mechanism **230** may be similar to what is described above. For the pet food pod **100'** shown in FIGS. **13** and **14**, the tab engaging element **244** component of the lid removal mechanism **230** may be an L-shaped hook. The orientation of the L-shaped hook may be crosswise relative to the hook **124'** on the underside of tab **122'**. In other words, the orientation may be as shown in FIG. **13**, in which the straight lower leg **245** (i.e., the portion **245**) of the L-shaped hook (i.e., of tab engaging element **244**) is shown end-on.

[0078] As the tab engaging element **244** moves towards the pet food pod **100'**, the lower leg **245** of the L-shaped hook may pass between the hook **124'** and the retainer **246** (the latter being immediately above pod retention element **116'**). Upon contacting and riding up the inwardly and upwardly sloped lateral face **117'** of nub **114'**, the lower leg **245** of tab engaging element **244** may press upwardly against the underside of the tab **122'**. This may initiate a prying away of the tab **122'** from the nub **114'** and generally away from container portion **110'**. Further translation of the tab engaging element **244** may cause the tab **122'** to fold over the container portion **110'**. Folding of the tab **122'** may cause the hook **124'** to become inverted with its tip pointing to the left in FIG. **13**. At that stage, further translation of the tab engaging element **244** along its trajectory may cause the lower leg **245** of the tab engaging element **244** to engage with hook **124'**. Further translation of the tab engaging element **244** to the right of FIG. **13** may pull the hook **124'** and tab **122'** along with it, peeling the lid **120'** away from the container portion **110'** in a similar manner to what was earlier described in connection with FIGS. **4-9**.

[0079] When the removed lid **120'** is above the lid disposal tray **248**, it may be dropped therein. In some embodiments, the tab engaging element **244** may be retractable. Retraction may cause the hook **124'** to slide off the lower leg **245**. In some embodiments, a fixed member within the body **202** can push the lid **120'** off the tab engaging element **244**.

[0080] After the lid **120'** has been removed from the container portion **110'** and dropped into the lid disposal tray **248**, the now opened pet food pod **100'** may be conveyed from the lid removal bay **210** to the feeding bay **270** to be presented to the pet for feeding. The conveying mechanism used to convey the pet food pod **100** to the feeding bay **270** may be similar to, or different from, what was used to convey the pet food pod **100'** from the receiving bay **260** to the lid removal bay **210**.

[0081] During conveyance to the feeding bay **270**, the pet food pod **100'** need not necessarily be maintained in the same orientation that it had assumed in the lid removal bay **210**. Thus, the conveyor in some embodiments could convey the pet food pod **100'** with its base **113'** on a flat surface versus rails, which may permit rotation of the pet food pod **100'** to occur during conveyance.

[0082] The opened pet food pod **100'** may be left in the feeding bay **270** for a predetermined amount of time that is sufficient for a pet to feed. In some embodiments, the device **200** may generate a predetermined notification, e.g., an auditory notification, when the wet pet food **118** has been presented for consumption. A house pet may come to associate the notification with newly available food. This may encourage timely consumption of wet pet food while it is fresh.

[0083] After a predetermined feeding interval, the controller may activate the second conveyor **295** to convey the empty, or substantially empty, used pet food pod **100'** to the pod disposal chamber **300**. The pod disposal chamber **300** may have an associated compactor mechanism (not expressly depicted). The compactor mechanism may be used to push the at least substantially empty pod container portion through the opening **302** (FIG. **12**A) toward a rear area of the pod disposal chamber **300** and/or to compact the at least substantially empty pod container portion(s) to maximize available storage space therewithin. As the pet food pod **100'** is pushed rearwardly within the tray, it may become upwardly angled in view of the angled floor of the pod disposal tray **290**. The angle may be about 25 to 30 degrees to minimize spillage from any uneaten food.

[0084] The rim **112'** of the angled pet food pod **100'** may come into contact with a floor **293** of the pod disposal tray **290** and may become caught on one of the grooves **297** or ridges in the floor. The grooves **297** or ridges may be configured (e.g., the grooves may be made to have a predetermined depth and/or width) to so catch the rim **112'**. This may prevent or limit slipping of the pet food pod **100'** back towards a front of the pod disposal chamber **300** and/or pod disposal tray **290**.

[0085] The controller thereafter may activate the disinfecting light source **304** and/or fan within fan unit **306** for a set duration after initial disposal of a pet food pod **100'** and/or at various intervals thereafter, until the pod disposal tray **290** is emptied. The rationale for such activation of the fan may be to expedite drying of any food remnants to delay spoilage and limit malodor.

[0086] In some embodiments, the device **200** may have an associated software application (app). The app may communicate with the embedded controller, e.g., to provide user notification regarding the status of the device **200**. The app may also allow the user to customize a pet's feeding plan or to override the plan on demand. The app may also provide a purchasing tool for customers to order pods automatically once a feeding plan has been chosen.

[0087] In some embodiments, the device **200** may incorporate a scale at the feeding bay **270** to monitor food consumption by weight. The app may be operable to provide the pet owner with data regarding the timing and amount of wet pet food consumption over time, e.g., by tracking patterns of food consumption. The app may provide a user indication if the pattern has changed by a predetermined amount, which may be a warning sign of an underlying health condition of a pet.

[0088] The design of device **200** should be robust yet safe for pets. There should not be parts where pets can get stuck or injure themselves, but it should also be difficult for pets to knock over the device **200** or to remove the tower **250** to access the pods.

[0089] In some embodiments, the device **200** should be reliable enough to work on without human intervention for at least two to three days. This may include limiting the chance of failure from interference from pets and ability to "unjam" or reset itself in case of a failure. In the case of a failure that cannot be automatically resolved, some embodiment of device **200** should be operable to trigger a user notification, e.g., via the app.

[0090] Various alternative embodiments are possible.

[0091] In some embodiments, the tab engaging element may be something other than an L-shaped hook. For example, in some embodiments, the tab engaging element may be a ring designed to engage a complementary tab engagement facilitator portion (e.g., a hook or tooth) of the peripheral tab of the pet food pod lid. In some embodiments.

[0092] In some embodiments, the pod retention element may be something other than a downwardly and inwardly angled lip or a flat ledge. The characteristics of the pod retention element may be based on (e.g., may be complementary to) the characteristics of the retainer to be used to substantially immobilize the pet food pod during lid removal. For example, in some embodiments, the pod retention element may be a lip or flange projecting laterally from a base of the container portion of the pet food pod. For such embodiments, the retainer may be an opposed pair of horizontal grooves in respective walls on opposite sides of a track within which the lip or flange may be slidably received.

What is claimed is:

1. A device for automatically serving wet pet food, the device comprising:

a body;

a lid removal bay in the body, the lid removal bay configured to receive a pet food pod, the pet food pod including a container portion containing wet pet food, the container portion being generally bowl-shaped and having a rim, the pet food pod further including a lid peelably and hermetically sealed to the rim, the lid having a peripheral tab;

a tab engaging element configured to engage the peripheral tab;

a lid removal actuator in the body, the lid removal actuator operable to, upon receipt of the pet food pod within the lid removal bay, actuate the tab engaging element along a trajectory over the pet food pod so that the tab engaging element will engage and pull the peripheral tab across the container portion to cause the lid to peel away from the rim, resulting in an open container portion with exposed wet pet food; and

a retainer configured to substantially immobilize the pet food pod within the lid removal bay during lid removal by the tab engaging element.

2. The device of claim **1** further comprising:

a pod storage tower removably attached to the body, the pod storage tower for receiving a stack of the pet food pods.

3. The device of claim **2** wherein the pod storage tower includes a pod alignment feature for promoting a uniform orientation of the pet food pods in the stack received in the pod storage tower.

4. The device of claim **2** further comprising:

a receiving bay in the body below the pod storage tower, the receiving bay configured to receive a pet food pod from the pod storage tower; and

a pod lowering mechanism for lowering a lowermost pod from the pod storage tower into the receiving bay.

5. The device of claim **4** further comprising a conveyor operable to convey a pet food pod from the receiving bay to the lid removal bay.

6. The device of claim **5** wherein the conveyor substantially maintains an orientation of the pet food pod during conveyance.

7. The device of claim **5** wherein the conveyor comprises a pair of parallel rails, wherein the container portion of the pet food pod has a base with a pair of straight parallel grooves formed in an underside of the base, and wherein the conveyor is operable to slidably translate the pet food pod atop the pair of parallel rails with each of the grooves seated on a respective one of the rails.

8. The device of claim **1** further comprising:

a lid disposal tray within the body adjacent to the lid removal bay, the lid disposal tray being configured to hold a plurality of lids after removal of the lids from respective ones of the pet food pods.

9. The device of claim **1** further comprising:

a feeding bay in the body configured to receive the open container portion of a pet food pod that has been conveyed from the lid removal bay, the feeding bay including an opening in a top surface of the body, the opening being surrounded by an inwardly facing lip, the lip being configured to obstruct pod removal from the feeding bay without obstructing access to wet pet food within the container portion.

10. The device of claim **9** further comprising:

a pod disposal chamber configured to receive a plurality of pet food pod container portions, in a substantially empty state, conveyed from the feeding bay.

11. The device of claim **10** further comprising:

a compactor mechanism operable to compact the plurality of pet food pod container portions, in the substantially empty state, within the pod disposal chamber.

12. The device of claim **10**, wherein the pod disposal chamber includes a removable pod disposal tray having a pair of longitudinal curbs and a floor between the curbs recessed downwardly relative to the curbs.

13. The device of claim **10**, wherein the pod disposal chamber includes a removable pod disposal tray having a floor and a plurality of crosswise grooves or ridges on the floor, each of the crosswise grooves or ridges being configured to catch the rim of a respective one of the pet food pod container portions to discourage slippage of the pet food pod container portion towards a front of the pod disposal tray.

14. The device of claim **1** further comprising:

a fan within the body; and

a removable filter positioned in the airflow path of the fan, the removable filter configured to absorb or neutralize odors from air circulated by the fan, the filter being removably secured within the device.

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